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Regular paper

Joint throughput and coverage maximization for moving users by optimum UAV positioning in the presence of underlaid D2D communications

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ABSTRACT

A fundamental challenge in using unmanned aerial vehicles as a flight base station is establishing coverage and creating the best possible rate. Drone location determines the maximum coverage probability and available rate. The paper considers a moving UAV as an aerial base station for downlink users with Gauss–Markov random movement and receiving interference from two users. First, we consider several points randomly in the area covered by the drone to find the optimal position. Using Algorithm (1), we calculate the maximum radius that the UAV can cover. After that, the algorithm (2) determines the distance of users to random points, the probability of coverage, and the available rate. In the end, we calculate the optimal point of the drone based on Algorithm (3). The results of the simulation using MATLAB show that after finding the optimal position of the drone, 95 percent of the users are covered and 0.15 Gbps is the average available rate for downlink users. Compared to other methods, it has the highest value.

1. Introduction

Recently, UAVs have attracted attention due to their capabilities such as obstacle avoidance, mobility, 3D movement, and fast deployment compared to base stations (BSs). In addition, UAVs are currently used in other cases such as: regulatory, military, goods delivery, etc. [1–4].

Finding the ideal position of UAVs is one of the essential challenges in UAV-based wireless communications. For this purpose, the mobility and adjustable height of drones can be a appropriate tool for the efficient deployment of drones. In reality, optimal deployment is a key design when using UAVs for coverage and maximum capacity, public safety, smart cities, storage, and Internet of Things (IoT) applications [5]. Ideal deployment of UAVs may be a challenging assignment since it depends on numerous components such as the environment of deployment (for example: geographic area), the ground users position, and the characteristics of the UAV-to-ground channel, which itself is a function of the UAV's height. Unlike ground base stations, UAVs must consider the impact of altitude on the air-to-ground (A2G) channel characteristics. In addition, when deploying the UAV, the flight time and the time to discover the optimal height should be considered in real-time scenarios, since they specifically influence the coverage of users [6].

Users who are located in the area and are moving, need timely service. For example, when the ground base stations are not available, if the user wants to immediately benefit from the communication coverage by the drone, the drone must be able to provide the maximum coverage to the user in real time and in the shortest possible time [7].

The optimal deployment of a UAV swarm, also known as a flying ad hoc network (FANET), to achieve on-demand coverage of mobile ground targets (MGTs) is challenging in terms of controlling UAV mobility to maximize coverage while maintaining quality of service. Therefore, the authors in [8] propose a joint topology control and routing (JTCR) protocol comprising three modules to perform crowd surveillance. Another challenge in deploying UAVs that can lead to a reduction in the available rate to terrestrial users is the user movement model, the users' movement speed and the number of users deployed in the area. For example, users moving too fast in the area can cause a decrease in packet delivery ratio (PDR), an increase in end-to-end delay, and an increase in the number of dropped packets (DP). Or in other cases, increasing the number of users resident in the area, regardless of the number of drones, can lead to a decrease in the rate available to users and a decrease in PDR [9–13].

In [14], the authors first raise the problem of the three-dimensional deployment of UAVs based on the global positioning system (GPS) to meet Quality of Service (QoS) requirements for different types of user distribution in the presence of interference and increased throughput. They then return their maximum value using a non-convex optimization problem. Finally, the best 3D position of the UAV is obtained using

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the reduced coordinate block method and solving the optimization problem.

In [15], the author covers two types of users by using trajectory optimization for 4 UAVs: users with group movement and the other group, users with random walking movement. For this purpose, first, an optimization problem with the aim of increasing the number of serviced users and includes 6 constraints is raised. Then it solves the optimization problem by using Lagrange coefficients, stochastic gradient descent (SGD) to find the minimum value and a heuristic algorithm.

A new supplementary UAV is proposed to be deployed inside the warning area to achieve a better monitoring quality. In [16], the authors identify the best location of the supplementary UAV by minimizing the maximum service distance in an acute triangle. In [17], the authors present a new framework to design the path of UAVs based on the prediction of users' movement information and the machine learning method. First, the authors present a joint path design and power control optimization problem to maximize the total instantaneous transmission rate while providing the required rate of users covered and moving by multiple UAVs. Then, using the multi-agent Q-learning method, they determine the optimal position of drones based on the users' origin location. The drone collects users' location information based on a set of real data extracted from the GPS. Finally, to predict the future situations of the users, they use the echo state network (ESN) method based on real data sets.

In [18–20], the authors present a replacement approach with the aim of 3D deployment of the minimum number of UAVs while maintaining the proposed quality. For this purpose, first the UAV positioning problem during a three-dimensional space is formulated as a mixed integer problem (MIP) optimization. Then, to realize an efficient solution, they present an innovative positioning algorithm which will easily adapt to the movement of users with a random walk model in a specific area and in the network. At the identical time, this algorithm increases the operational capacity and service of users. In [21], a multi-UAV enabled IoT is proposed, where the UAVs as base stations send information to the ground IoT nodes via downlink within the flight time. And a fair energy-efficient resource optimization scheme for the IoT is studied to ensure fair energy consumption of multiple UAVs.

In [22], a multi-UAV enabled mobile Internet of Vehicles (IoV) model is proposed, where the UAVs track to serve the mobile vehicles and send downlink information to the vehicles during the flight time. Considering the constraints of anti-collision and communication interference between the UAVs, the system throughput is maximized by jointly optimizing vehicle communication scheduling, UAV power allocation and UAV trajectory.

In [23], the authors consider an UAV relay communication system assisted by a reconfigurable intelligent surface (RIS), where the RIS is mounted on the UAV and can move at high speed. In fact, the authors maximize the average downlink throughput by jointly optimizing the UAV trajectory, passive RIS beamforming and resource power allocation for each time slot.

In [24], the authors consider the spectrum sharing scheduling problem for full-duplex UAV relay systems with D2D communications, where a mobile UAV contributes to the communication as a full-duplex relay. The authors' design aims to maximize the total output under the transmission power budget, while ensuring coexistence with terrestrial D2D pairs, satisfying information causality and UAV trajectory constraints.

In [25], the authors investigate the resource allocation problem for UAV-assisted networks, where a UAV acting as an energy source allocates radio frequency energy for energy-abundant D2D. Information to be transmitted The objective is to maximize the average power over a time horizon while preserving the energy causality constraint under a generalized capture, transfer, storage model, which leads to a non-convex problem.

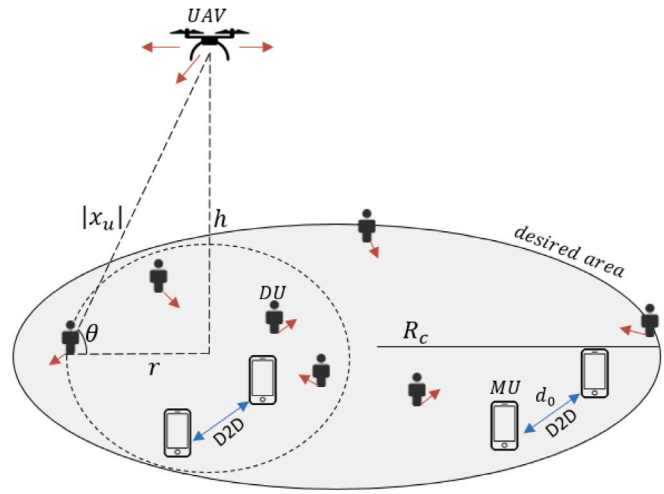

Fig. 1. System model.

The main goal in this paper is to increase the coverage probability and available rate for DUs with Gauss–Markov random movement compared to other methods. The assumptions and new ideas of the problem are as follows:

In this paper, firstly, we consider a moving UAV that serves two types of users, including downlink users and D2D transmitter–receiver pairs with Rayleigh fading communication channel. At the same time, D2D transmitters cause interference to the receivers. Downlink users have Gauss–Markov random mobility and pairs of D2D users are distributed by Poisson point process (PPP) in the circular region. We note that the final position of both types of users is according to the polar coordinate system. First, by using algorithm 1, we obtain the maximum radius of the area covered by the drone. This is done by setting a threshold for the average coverage probability for downlink users and assigning a value for the radius of the coverage area per altitude from 1 to 4500 m. After calculating the maximum radius, using a completely new method based on GPS and an innovative algorithm in non-real time, we calculate the optimal position of the drone. In the mentioned method, the drone already knows the position of the users according to the Cartesian coordinate system. For this, we first consider a number of random points with a uniform distribution in the circular area. Then, using algorithm 2, we calculate the distance of users to any random point for all time slots. After calculating the distances, the algorithm checks whether the distance is less than the maximum radius of the UAV coverage area R_{max} or not. If the distance of the random point to the user is less than the radius, the user will not be covered, otherwise it will be covered. In the next step, the number of downlink users who have been covered for all random points and for all time slots are obtained (algorithm 2). Finally, using Algorithm 3, the optimal position of the UAV is obtained in all the time slots that have the highest coverage probability and available rate for downlink users. The optimal position obtained by using Algorithm 3 provides a higher coverage probability and accessible rate for DUs. The remainder of this paper is organized as follows. In Section 2, the system model, the proposed method to find the optimal position of the UAV and the algorithms are presented. In Section 3, the simulation results are stated and in Section 4, the conclusion of the article is given.

2. System model

The model used in this article follows [26]. In this model, we consider a circular area of dense urban type with radius R_c. In this area, there are two types of users including pairs of D2D users (transmitter–receiver) with density λ_d and downlink users with Gauss–Markov random motion with density λ_{du} and are covered by a low altitude platform

(LAP) UAV. Since there is communication from one point (drone) to several points (users), we have considered the antenna of the drone to be omnidirectional with a 360 degree radiation angle. Fig. 1 shows the system model of the article. In Fig. 1, h is the altitude of the drone to the ground, r is the radius of the area covered by the drone, θ is the elevation angle of the drone, $|X_u| = \sqrt{h^2 + r^2}$ is the distance between the drone and a user, and d_0 is the fixed distance between a The transmitter–receiver pair is D2D. Eqs. (1) and (2) show the signal-to-interference-plus-noise ratio (SINR) for a downlink receiver and a D2D receiver, respectively.

$$\gamma_u = \frac{P_{r,u}}{I_d + N} \tag{1}$$

$$\gamma_d = \frac{P_{r,d}}{I_d^c + I_u + N} \tag{2}$$

Where, $P_{r,u}$ signal power received from UAV, $P_{r,d}$ signal power received from D2D transmitter, I_d total interference power from D2D transmitters, N noise power, I_d^c total interference from other users D2D, I_u are the interference received from the drone. Note that the formulated SINR value is calculated for only one receiver. If we want to get the SINR value for all users, we get the mathematical expectation $\mathbb{E}(.)$ from the SINR. The reason for using mathematical hope is that the noise and interference parameters are random and statistical parameters, and for averaging, mathematical hope should be used. Averaging is included in the calculation of the average coverage probability and available rate. In the circular area, there are three types of interference, including interference from the drone side for D2D users, interference from other D2D transmitters for the D2D receiver, and interference from all D2D transmitters for downlink users. Eqs. (3), (4) and (5) show the power of the signal received from the UAV, the total amount of interference received from all other D2D transmitters, and the interference received from the UAV, respectively.

$$P_{r,d} = p_d d_0^{-\alpha_d} g_0 \tag{3}$$

$$I_d^c = \sum_{i \neq 0} p_d d_i^{-\alpha_d} g_i \tag{4}$$

$$I_d = \sum_{i} p_d d_i^{-\alpha_d} g_i \tag{5}$$

Where, p_d is the transmission power of the D2D transmitter, α_d is the path loss power between D2D users, g_0 is the channel gain between a D2D receiver and its corresponding transmitter, g_i is the channel gain of the ith D2D transmitter, and d_i is the distance between a receiver and the ith D2D transmitter. The index $i = 0$ is used for the selected D2D transmitter–receiver pair. We note that all the parameters for the D2D transmitter–receiver pair are considered based on the Rayleigh fading channel model.

The distribution of D2D users on the circular area is uniform, and the channel model between UAV and downlink users is of A2G type, taking into account the dominant signals, namely, line of sight (LoS) and none line of sight (NLoS). Also, the channel model between pairs of D2D users is Rayleigh fading type. The probability of LoS and NLoS are obtained from (6) and (7), respectively.

$$P_{LoS} = \frac{1}{1 + Ce^{-B[\theta - C]}} \tag{6}$$

$$P_{NLoS} = 1 - P_{LoS} \tag{7}$$

Where, C and B are constant values that depend on the type of environment (rural, urban, dense urban or other). The value of θ is obtained from (8).

$$\theta = \frac{180}{\pi} \times \sin^{-1}\left(\frac{h}{|X_u|}\right) \tag{8}$$

For each of the LOS or NLOS connection links between the user and the UAV, the signal received from the UAV can be calculated by (9).

$$P_{r,u} = \begin{cases} P_u |X_u|^{-\alpha_u} & \text{LoS Link} \\ \eta P_u |X_u|^{-\alpha_u} & \text{NLoS Link} \end{cases} \tag{9}$$

Where, P_u is the transmitted power of the UAV, α_u is the path loss of the user-UAV link and η is an additional attenuation factor for the NLoS link. In this article, our main focus is on downlink users.

For a DU located at position (r, φ), the coverage probability in the absence of D2D users is given by (10) [26].

$$\begin{aligned} P_{cov,du}(r,\varphi,\beta) &= P[\gamma_u \geq \beta] = P_{LoS}(r) P[\gamma_u \geq \beta \mid LoS] \\ &+ P_{NLoS}(r) P[\gamma_u \geq \beta \mid NLoS] = P_{LoS}(r) I\left[r \leq \left(\frac{P_u}{\beta N}\right)^{\frac{1}{\alpha_u}}\right] \\ &+ P_{NLoS}(r) I\left[r \leq \left(\frac{\eta P_u}{\beta N}\right)^{\frac{1}{\alpha_u}}\right] \end{aligned} \tag{10}$$

The above relationship shows the coverage probability for a single downlink user. By taking the mathematical expectation (as mentioned earlier) of $P_{cov,du}(r,\varphi,\beta)$ on the cell and with the radius R_c, the average coverage probability for downlink users assuming a small number of D2D users (for preventing from interference) is obtained as (11).

$$\begin{aligned} \bar{P}_{cov,du}(r,\varphi,\beta) &= \mathbb{E}_{r,\varphi}\left[P_{cov,du}(r,\varphi,\beta)\right] \\ &= \int_0^{min\left[\left(\frac{P_u}{\beta}\right)^{\frac{1}{\alpha_u}}, R_c\right]} P_{LoS}(r) \frac{2r}{R_c^2}\, dr \\ &+ \int_0^{min\left[\left(\frac{\eta P_u}{\beta}\right)^{\frac{1}{\alpha_u}}, R_c\right]} P_{NLoS}(r) \frac{2r}{R_c^2}\, dr \end{aligned} \tag{11}$$

Where, (r, φ) is the position of the downlink user in terms of the polar coordinate system in the circular area and β are the SINR threshold. If the interference of D2D users is insignificant compared to the UAV due to the density and low transmission power, it is an accurate expression for the coverage probability of downlink users. Therefore, when the signal received from the UAV is dominant compared to the interference from the D2D transmitters, it represents the maximum performance achievable for downlink users.

2.1. Gauss–Markov mobility model

The movement of downlink users follows the GM random movement model [9]. This model is used to simulate Ad-Hoc networks. Figs. 2–4 show the movement of the first, second and third user based on Gauss–Markov random movement in 50 time slots, respectively. This figures are based on the movement interval $x = [1, 1000]$ and $y = [1, 1000]$.

In Fig. 2, each DU is initially assigned the current speed and direction, then, at specified time intervals n, movement occurs by updating the speed and direction. In particular, the value of speed and direction in the nth stage are obtained based on the value of speed and direction in sample $(n-1)$ and a random variable using (12) and (13).

$$v_{u_i}[n] = \alpha v_{u_i}[n-1] + (1-\alpha)\bar{v} + \sqrt{\left(1-\alpha^2\right) v_{x_{u_i}}[n-1]} \tag{12}$$

$$d_{u_i}[n] = \alpha d_{u_i}[n-1] + (1-\alpha)\bar{d} + \sqrt{\left(1-\alpha^2\right) d_{x_{u_i}}[n-1]} \tag{13}$$

Where, $v_{u_i}[n]$ and $d_{u_i}[n]$ are the new speed and direction for the ith downlink user in the nth time slot, respectively. α is the tuning parameter used for random variation. $\bar{v}$ and $\bar{d}$ are constants that represent the average value of speed and direction, respectively. $v_{x_{u_i}}[n-1]$ and $d_{x_{u_i}}[n-1]$ are random variables and are selected from a Gaussian (normal) distribution. Random average levels are created by setting the value of α between zero and one. We note that the value of α in the simulation of Figs. 2,3 and 4, for each time slot, is created randomly and automatically in the range of zero and one. In each interval, the

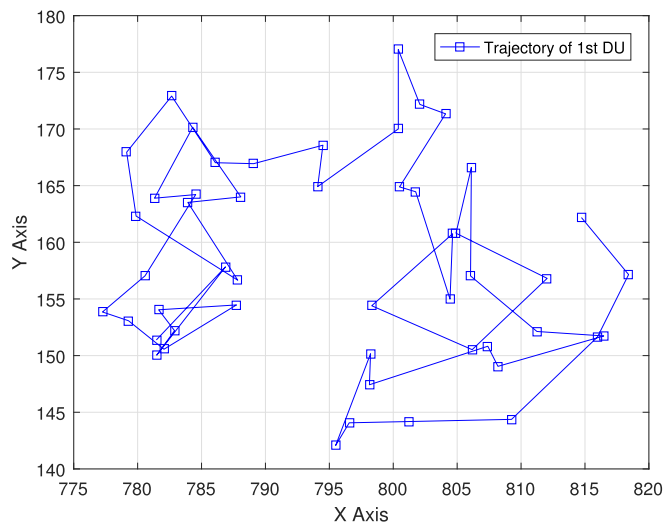

Fig. 2. The trajectory of the first DU for 50 time slots (α value is randomly chosen between 0 and 1).

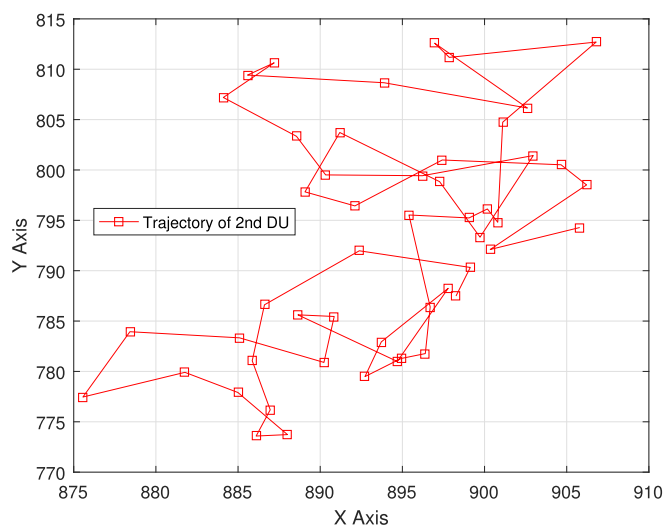

Fig. 3. The trajectory of the second DU for 50 time slots (α value is randomly chosen between 0 and 1).

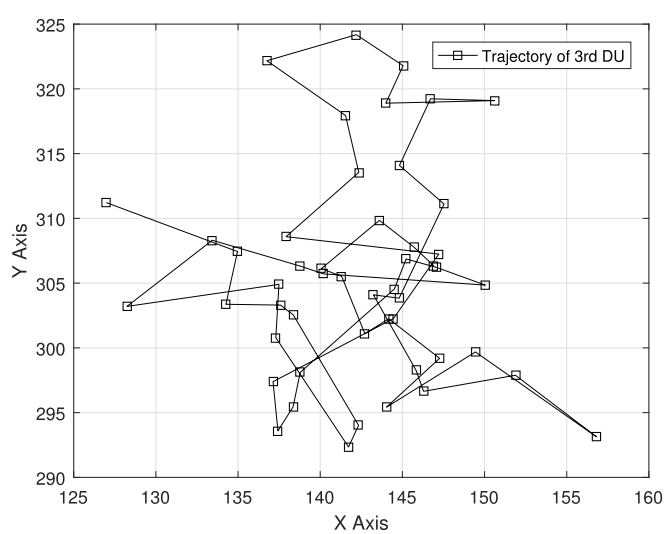

Fig. 4. The trajectory of the third DU for 50 time slots (α value is randomly chosen between 0 and 1).

next location is calculated based on the current location, speed and direction of movement. A summary of the state of setting the parameter α is obtained in the form of (14).

$$\alpha = \begin{cases} 0 & \text{Brownian Motion} \\ 1 & \text{Linear Motion} \end{cases} \tag{14}$$

In particular, in the nth time slot, the position of downlink users in terms of the Cartesian coordinate system is described by (15) and (16).

$$x_{u_i}[n] = x_{u_i}[n-1] + \left(v_{u_i}[n-1] \times \cos(d_{u_i}[n-1])\right) \tag{15}$$

$$y_{u_i}[n] = y_{u_i}[n-1] + \left(v_{u_i}[n-1] \times \sin(d_{u_i}[n-1])\right) \tag{16}$$

Where, $(x_{u_i}[n], y_{u_i}[n])$ and $(x_{u_i}[n-1], y_{u_i}[n-1])$ are the x and y coordinates of the ith position of the DU in time slots n and n-1, respectively. Also, $v_{u_i}[n-1]$ and $d_{u_i}[n-1]$ are the speed and direction of the downlink user in time slot $n-1$, respectively. Here, time slot means dividing time into different parts. If T_{uav} is the total flight time of the UAV, the time division will be $\frac{T_{uav}}{n}$. The Gauss–Markov random mobility model has the highest PDR and the lowest End-to-End delay among other motion models [10,11].

Two key points can be extracted from [10,11]. First, the speed of user mobility has reduced the PDR and increased the end-to-end delay. Second, the PDR rate and End-to-End delay for the GM mobility model are higher and lower, respectively. Therefore, this model is chosen as the mobility model of this article.

In our method, downlink users first move by Gauss–Markov random movement model in each time slot, and their movement position is stored in the UAV memory according to the Cartesian coordinate system. Then, the UAV calculates the optimal position of the UAV for each time slot using the positions stored in the UAV's memory, in a non-real-time manner and using Algorithms 2 and 3.

2.2. Proposed method to find the optimum position of the UAV

In this method, it is assumed that the UAV knows the location of the users in each time slot in advance. In other words, the information about the location of downlink users is available with high accuracy using GPS. According to the type of environment, the ith position of the downlink user and the D2D user pair for the nth time slot and based on the Cartesian coordinate system are obtained according to (17) and (18).

$$P_{u_i}^{du,cart}[n] = \left(x_{u_i}^{du,cart}[n], y_{u_i}^{du,cart}[n]\right), i = 1, \dots, n_{du}, n = 1, \dots, M \tag{17}$$

$$P_{u_i}^{d2d,cart}[n] = \left(x_{u_i}^{d2d,cart}[n], y_{u_i}^{d2d,cart}[n]\right), i = 1, \dots, n_{d}, n = 1, \dots, M \tag{18}$$

Eqs. (17) and (18) show the position of the ith downlink user and the ith D2D user pair for the nth time slot, respectively. Users' position is converted from Cartesian coordinate system to polar by (19), (20), (21) and (22).

$$P_{u_i}^{du,pol}[n] \rightarrow r_{u_i}^{du,pol}[n] = \sqrt{\left(x_{u_i}^{du,cart}[n]\right)^2 + \left(y_{u_i}^{du,cart}[n]\right)^2} \tag{19}$$

$$P_{u_i}^{d2d,pol}[n] \rightarrow r_{u_i}^{d2d,pol}[n] = \sqrt{\left(x_{u_i}^{d2d,cart}[n]\right)^2 + \left(y_{u_i}^{d2d,cart}[n]\right)^2} \tag{20}$$

$$P_{u_i}^{du,pol}[n] \rightarrow \varphi_{u_i}^{du,pol}[n] = \begin{cases} \tan^{-1}\left(\frac{y_{u_i}^{du,cart}[n]}{x_{u_i}^{du,cart}[n]}\right) & x_{u_i}^{du,cart}[n] > 0 \\ \tan^{-1}\left(\frac{y_{u_i}^{du,cart}[n]}{x_{u_i}^{du,cart}[n]}\right) + \pi & x_{u_i}^{du,cart}[n] < 0, y_{u_i}^{du,cart}[n] \geqslant 0 \\ \tan^{-1}\left(\frac{y_{u_i}^{du,cart}[n]}{x_{u_i}^{du,cart}[n]}\right) - \pi & x_{u_i}^{du,cart}[n] < 0, y_{u_i}^{du,cart}[n] < 0 \\ \frac{\pi}{2} & x_{u_i}^{du,cart}[n] = 0, y_{u_i}^{du,cart}[n] > 0 \\ -\frac{\pi}{2} & x_{u_i}^{du,cart}[n] = 0, y_{u_i}^{du,cart}[n] < 0 \end{cases} \tag{21}$$

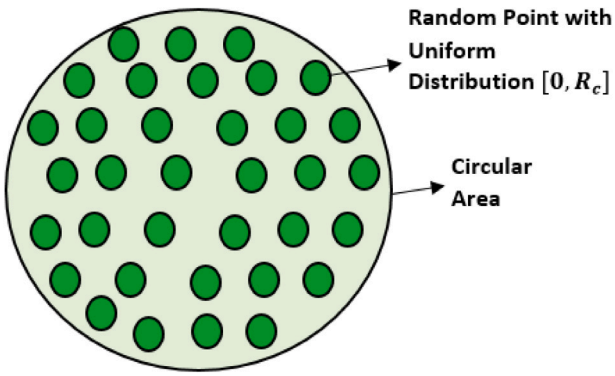

Fig. 5. Random points with a uniform distribution in the circular area.

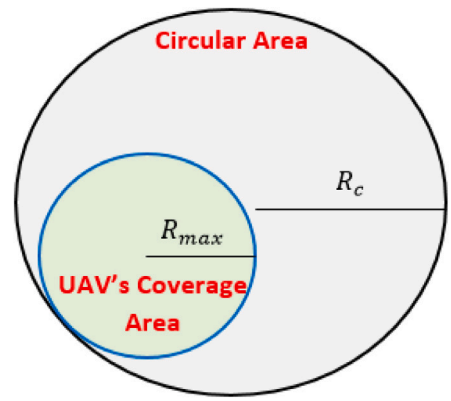

Fig. 6. UAV coverage radius in the circular area.

$$P_{u_i}^{d2d,pol}[n] \rightarrow \varphi_{u_i}^{d2d,pol}[n] = \begin{cases} \tan^{-1}\left(\frac{y_{u_i}^{d2d,cart}[n]}{x_{u_i}^{d2d,cart}[n]}\right) & x_{u_i}^{d2d,cart}[n] > 0 \\ \tan^{-1}\left(\frac{y_{u_i}^{d2d,cart}[n]}{x_{u_i}^{d2d,cart}[n]}\right) + \pi & x_{u_i}^{d2d,cart}[n] < 0, y_{u_i}^{d2d,cart}[n] \geqslant 0 \\ \tan^{-1}\left(\frac{y_{u_i}^{d2d,cart}[n]}{x_{u_i}^{d2d,cart}[n]}\right) - \pi & x_{u_i}^{d2d,cart}[n] < 0, y_{u_i}^{d2d,cart}[n] < 0 \\ \frac{\pi}{2} & x_{u_i}^{d2d,cart}[n] = 0, y_{u_i}^{d2d,cart}[n] > 0 \\ -\frac{\pi}{2} & x_{u_i}^{d2d,cart}[n] = 0, y_{u_i}^{d2d,cart}[n] < 0 \end{cases} \tag{22}$$

Finally, by integrating (19), (20), (21) and (22), the positions of downlink and D2D users in terms of polar coordinates are obtained as (23) and (24), respectively.

$$P_{u_i}^{du,pol}[n] = \left(r_{u_i}^{du,pol}[n], \varphi_{u_i}^{du,pol}[n]\right), i = 1, \ldots, n_{du}, n = 1, \ldots, M \tag{23}$$

$$P_{u_i}^{d2d,pol}[n] = \left(r_{u_i}^{d2d,pol}[n], \varphi_{u_i}^{d2d,pol}[n]\right), i = 1, \ldots, n_d, n = 1, \ldots, M \tag{24}$$

In our method, to find the optimal position of the drone, first, we consider the j points as $d_{rp} = \{d_{rp_1}, d_{rp_2}, \ldots, d_{rp_j}\}$ and completely random in the circular area with radius R_c. Where, $j = nv$ is the number of random points. These points are randomly distributed and generated only once by the UAV processor and used for all time slots. Fig. 5 shows the position of random points in terms of the polar coordinate system in the circular area. We note that, these points are fixed for all time slots and there is no need to reproduce points for each time slot. The distribution of points (d_1 to d_j) in the circular area is described by Eq. (25).

$$f_{R_{rp},\phi_{rp}}\left(r_{rp}, \varphi_{rp}\right) = \frac{r_{rp}}{A_{ca}}, 0 < r_{rp} < R_c, 0 < \varphi_{rp} < 2\pi \tag{25}$$

Proof. See Appendix A. □

In (25), r_{rp} and φ_{rp} are the radius and angle of random points in the polar coordinate system, respectively and $A_{ca} = \pi R_c^2$ is the area of the circular area. More precisely, $f_{R_{rp},\phi_{rp}}\left(r_{rp}, \varphi_{rp}\right)$ describes the distribution of users uniformly over the surface area assumed to be a circle here. The values of r_{rp} and φ_{rp} are obtained from (26) and (27), respectively.

$$r_{rp} = \{r_{rp_1}, r_{rp_2}, \ldots, r_{rp_j}\} \tag{26}$$

$$\varphi_{rp} = \{\varphi_{rp_1}, \varphi_{rp_2}, \ldots, \varphi_{rp_j}\} \tag{27}$$

By integrating d_{rp} and (26) and (26), (28) is obtained.

$$d_{rp} = \left\{\underbrace{\left(r_{rp_1}, \varphi_{rp_1}\right)}_{d_{rp_1}}, \underbrace{\left(r_{rp_2}, \varphi_{rp_2}\right)}_{d_{rp_2}}, \ldots, \underbrace{\left(r_{rp_j}, \varphi_{rp_j}\right)}_{d_{rp_j}}\right\} \tag{28}$$

2.2.1. Calculation of the maximum radius of the area covered by the UAV

After creating the points on the circular area, two factors including the probability of coverage and the available rate for downlink users should be calculated for all the points. For this, we first calculate the maximum radius that the drone can cover (R_{max}). The coverage radius is defined as the maximum radius where the probability of coverage for all DUs (located in the coverage area) is greater than a specified threshold. In this case, the UAV meets the coverage requirement of every downlink user that is within its coverage area. In other words, the maximum radius of the drone determines how many users out of the total number of users are covered by the drone. Fig. 6 shows a more detailed view of the maximum coverage radius of the UAV in the circular area. The maximum coverage radius for a UAV at a height of h and with a transmitted power of P_u is defined by Eq. (29) [26].

$$R_{max} = max\left\{R \mid P_{cov,du}(\beta, R) \geqslant \varepsilon, P_u, h\right\} = P_{cov,du}^{-1}(\beta, \varepsilon) \tag{29}$$

Eq. (29) is the problem of finding the maximum value of R according to the threshold ε. To solve this problem, the radius of the drone R can be set in a certain interval and in any value where the coverage probability is greater than or equal to the threshold ε, R_{max} can be chosen. According to [27], the optimal altitude of the UAV h^* is a function of the average path loss. Therefore, we have:

$$\bar{L}(R, h) = \left[\left(P_{LoS} \times L_{LoS}\right) + \left(P_{NLoS} \times L_{NLoS}\right)\right] \tag{30}$$

Where, L_{LoS} and L_{NLoS} are the average path loss for LOS and NLOS, respectively. The values of these parameters are obtained from (31) and (32), respectively [27].

$$L_{LoS}(dB) = 20\log\left(\frac{4\pi f_c |X_u|}{c}\right) + \xi_{LoS} \tag{31}$$

$$L_{NLoS}(dB) = 20\log\left(\frac{4\pi f_c |X_u|}{c}\right) + \xi_{NLoS} \tag{32}$$

Where, ξ_{LoS} and ξ_{NLoS} are the average additional loss to the free-space propagation loss that depend on the type of environment, c is the speed of light and f_c is the carrier frequency. For a UAV that sends its signal with the power P_u, the received power is obtained in the form of (33).

$$p_{r,u}(dB) = P_u - \bar{L}(R, h) \tag{33}$$

A downlink user is covered by UAV, if we have:

$$\gamma_u(R, h) = \frac{p_{r,u}}{N + I} > \beta \tag{34}$$

Obviously, to find the maximum reachable coverage radius, we must have $\gamma_u(R, h) = \beta$. For a fixed transmit power, the optimal UAV altitude that results in maximum coverage is calculated by solving the following equation:

$$\frac{180\left(\xi_{LoS} - \xi_{NLoS}\right) BZ}{\pi (Z+1)^2} - \frac{20\mu}{\log(10)} = 0 \tag{35}$$

Where, $Z = Ce^{-B\left[\frac{180}{\pi}\tan^{-1}(\mu)-C\right]}$ and $\mu = \frac{h}{R}$. By solving (30), $\mu^* = \frac{h^*}{R_{max}}$ is calculated and using (33), h^* is obtained. Now, we assume that the target area to be covered is fixed with radius R_c and the objectives are set to find an optimal altitude where the minimum transmit power is required to cover the target area. So we have [27]:

$$\frac{\partial P_u}{\partial h} = \frac{\partial \bar{L}\left(R_c, h\right)}{\partial h} = 0 \rightarrow h^* = \mu^* R_c \tag{36}$$

Now, according to the value calculated for the optimal altitude in (36), we can include the optimal value to reduce the transmitted power in the calculation of R_{max}.

According to (36) and assuming that the value of R is constant, we can obtain the optimal height of the UAV for n time slots. So we have:

$$h^*[n] = \frac{\partial P_u}{\partial h[n]} = \frac{\partial \bar{L}\left(R[n], h[n]\right)}{\partial h[n]}, R = constant \tag{37}$$

We note that here the constant R means that the maximum radius of the area covered by the drone (R_{max}) is constant for all time slots. For R with constant value in n time slots from 1 to M, we have:

$$R[1] = R[2] = R[3] = \cdots = R[M] \tag{38}$$

According to (37) and (38), by calculating the optimal height in one of time slots 1 to M, we can obtain the optimal height of the UAV for all time slots. So we have:

$$h^*[n] = \frac{\partial \bar{L}\left(R[n], h[n]\right)}{\partial h[n]} = \left[\left(\left(S_1 \times S_2\right) + S_3\right) - \left(\left(S_4 \times S_5 \times S_6\right) + S_7\right)\right] = 0 \tag{39}$$

In (39), S_1 to S_7 are obtained from relations (40) to (46), respectively.

$$S_1 = \frac{1}{\left(1 + Ce^{-B\left(180\sin^{-1}\left(\frac{\frac{h}{\sqrt{h^2+r^2}}}{\pi} - c\right)\right)}\right)^2} \times e^{-B\left(\frac{180\sin^{-1}\left(\frac{h}{\sqrt{h^2+r^2}}\right)}{\pi} - c\right)} \tag{40}$$

$$S_2 = 180\left(\frac{\frac{20\ln 4\pi f_c\sqrt{h^2+r^2}}{c}}{\ln(2)} + \xi_{LoS}\right)CB \times \left(\frac{1}{\sqrt{h^2+r^2}} - \frac{h^2}{(h^2+r^2)^{\frac{3}{2}}}\right) \tag{41}$$

$$S_3 = \frac{20h}{\left(1 + Ce^{-B\left(\frac{180\sin^{-1}\left(\frac{h}{\sqrt{h^2+r^2}}\right)}{\pi} - C\right)}\right)(h^2+r^2)\ln(2)} \tag{42}$$

$$S_4 = \frac{1}{\left(1 + Ce^{-B\left(180\sin^{-1}\left(\frac{\frac{h}{\sqrt{h^2+r^2}}}{\pi} - C\right)\right)}\right)\pi\sqrt{-\frac{h^2}{h^2+r^2} + 1}} \tag{43}$$

$$S_5 = 180CB\left(\frac{1}{\sqrt{h^2+r^2}} - \frac{h^2}{(h^2+r^2)^{\frac{3}{2}}}\right)e^{-B\left(180\sin^{-1}\left(\frac{\frac{h}{\sqrt{h^2+r^2}}}{\pi} - C\right)\right)} \tag{44}$$

$$S_6 = \left(\frac{\frac{20\ln\left(4\pi f_c\sqrt{h^2+r^2}\right)}{C}}{\ln(2)} + \xi_{NLoS}\right) \tag{45}$$

$$S_7 = \frac{20h\left(1 - \frac{1}{Ce^{-B\left(180\sin^{-1}\left(\frac{\frac{h}{\sqrt{h^2+r^2}}}{\pi} - C\right)\right)}}\right)}{(h^2+r^2)\ln(2)} \tag{46}$$

Algorithm 1 shows a solution to find R_{max}.

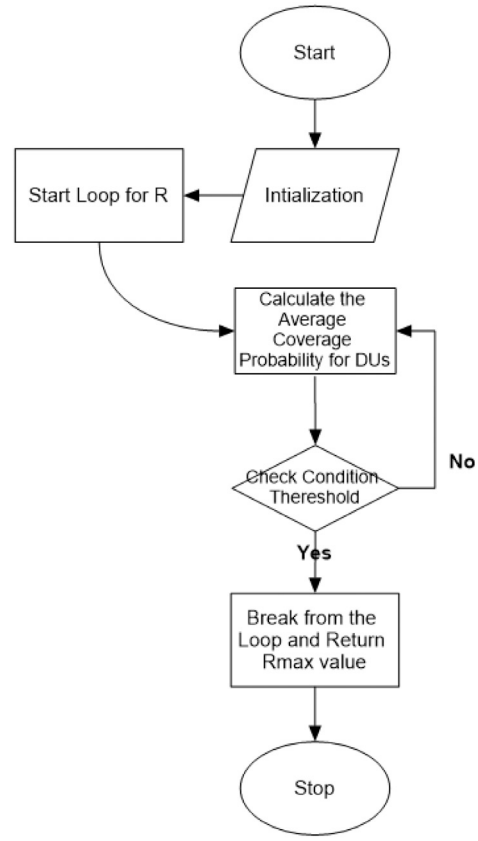

Fig. 7. Flowchart of Algorithm 1.

Algorithm 1 Calculation of the maximum value of the radius of the area covered by the UAV

1st step: set parameters $\beta, R^L, R^L, \varepsilon, h^*$
2nd step: start the loop for setting the lower and upper bounds for the R parameter
3rd step: calculate the value of $P_{cov,du}[R, h^*, \beta]$
4th step: check the condition $P_{cov,du} > \varepsilon \parallel P_{cov,du} = \varepsilon$
5th step: the condition is true → exit from loop and return the value of R_{max}
6th step: the condition is false → return to the loop (Second step)

In the first step of Algorithm 1, the parameters of the optimal altitude h^*, the lower bound of the radius R^L, the upper bound of the radius R^u, the signal-to-interference plus the noise ratio threshold β and the threshold for finding the maximum radius ε are set. In the second and third steps, a loop for the radius of the UAV coverage area with a minimum and maximum value is defined to test different values for the coverage probability of downlink users, i.e. $P_{cov,du}[R, h^*, \beta]$. In the fourth step, the algorithm checks that the calculated coverage probability is greater than or equal to the threshold ε. In the fifth step, if the condition is true, the current value of R is selected as R_{max}, and the loop ends, and the algorithm returns the maximum value of the radius of the area covered by the UAV. In the sixth step, if the condition is not true, the loop again calculates the coverage probability for the new value of R. In this algorithm, the loop will repeat until the threshold condition ε is not met (greater than or equal), up to the upper bound value of R. Fig. 7 shows the flowchart of Algorithm 1.

2.2.2. Measurement of coverage probability and available rate of DUs for all random points

To do this, we first obtain the distance of users to random points for all time slots. The distance of the ith user to the jth random point, which is the center of the area covered by the UAV with the radius R_{max} in the polar coordinate system, is obtained by (47).

$$d_{u_i,d_j}[n] = \sqrt{r_{u_i}[n] + r_{rp_j}[n] - 2r_{u_i}[n]r_{rp_j}[n]\cos(\varphi_{u_i}[n] - \varphi_{rp_j}[n])} \tag{47}$$

Where, $r_{u_i}[n]$ radius of the ith user, $r_{rp_j}[n]$ radius of the jth random point, $\varphi_{u_i}[n]$ angle of the ith user and $\varphi_{rp_j}[n]$ angle jth is the random point in the polar coordinate system. Also, as we said before, n represents the time slot. A user will be covered by the drone if the user's distance to the random point (the center of the drone's coverage area) is less than R_{max}. Fig. 8 shows the details of the driving problem. In Fig. 8, the number of pairs of D2D users is assumed to be small, the coverage or non-coverage of DUs can be determined by two states of zero and one. Zero means no coverage and one means full coverage of the user by the drone. We note that the coverage or lack of coverage of the downlink user has a direct relationship with the number of D2D transmitters. If the number of D2D transmitters is large, the receiver will receive a lot of interference from the D2D transmitter, so in this method, we consider the number of pairs of D2D users to be low so that the effect of interference can be neutralized.

Eqs. (48) and (49) show the lower bound and the upper bound of the average coverage probability for DUs in the desired area and in the presence of D2D users, respectively [26].

$$\begin{aligned}\bar{P}^{L}_{cov,du}(\beta,h) = &\int_0^{R_c} P_{LoS}(r,h)L_I\left(\frac{P_u|X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{{R_c}^2}\,dr \\ &+ \int_0^{R_c} P_{NLoS}(r,h)L_I\left(\frac{\eta P_u|X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{{R_c}^2}\,dr\end{aligned} \tag{48}$$

$$\begin{aligned}\bar{P}^{U}_{cov,du}(\beta,h) = &\int_0^{R_c} P_{LoS}(r,h)U_I\left(\frac{P_u|X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{{R_c}^2}\,dr \\ &+ \int_0^{R_c} P_{NLoS}(r,h)U_I\left(\frac{\eta P_u|X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{{R_c}^2}\,dr\end{aligned} \tag{49}$$

Where, $\beta N \leqslant P_u\|X_u\|^{-\alpha_u}$ and for each $T > 0$, we have [26]:

$$\begin{aligned}L_I(T) = &\left[1 - \frac{2\pi\lambda_d\Gamma\left(1+\frac{2}{\alpha_d}\right)}{\alpha_d - 2}\left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}}\right] \\ &\times exp\left(-\pi\lambda_d\left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}}\Gamma\left(1+\frac{2}{\alpha_d}\right)\right)\end{aligned} \tag{50}$$

$$U_I(T) = exp\left(-\pi\lambda_d\left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}}\Gamma\left(1+\frac{2}{\alpha_d}\right)\right) \tag{51}$$

Where, $\lambda_d = \frac{n_d}{\left(\pi R_c^2\right)}$ and $\lambda_{du} = \frac{n_{du}}{\left(\pi R_c^2\right)}$ are the density of D2D and downlink users in the cell, respectively. Also, the gamma function $\Gamma(.)$ is defined as (52).

$$\Gamma(t) = \int_0^{\infty} x^{t-1}e^{-x}\,dx \tag{52}$$

Proof. See Appendix B. □

The status of the ith user at a distance from the jth random point and for the nth time slot is obtained as follows:

$$Z_{u_i,d_{rp_j}}[n] = \begin{cases} 1 & \text{Covered} \\ 0 & \text{Uncovered}\end{cases} \tag{53}$$

$n = 1,\ldots,M, i = 1,\ldots,n_{du}, i \neq j$

Where, $Z_{u_i,d_{rp_j}}[n]$ specifies the status of the ith user at a distance from the jth random point and for the nth time slot. By counting the number of users for each random point, the total number of users covered for that point can be calculated. If $N_{us,d_{rp_j}}[n]$ is the total number of users served by the jth random point and in the nth time slot, the coverage probability for the users of the radio frequency is obtained by (54).

$$P_{du,d^{cov}_{rp_j}}[n] = \frac{N_{us,d_{rp_j}}[n]}{100} \tag{54}$$

To calculate the available rate for downlink users, we first obtain the channel capacity. The Shannon–Hartley theorem expresses the channel capacity $\bar{C}_{du}$, meaning the most compact theoretical upper bound of the data information rate at an arbitrarily low error rate using the average received signal power $p_{r,u}$ through a communication channel under N noise power or I interference can communicate. So, we have:

$$\bar{C}_{du} = W\log_2\left(1 + \frac{p_{r,u}}{N+I}\right) \tag{55}$$

Where, W is the bandwidth. By multiplying the channel capacity in (54), the available rate for downlink users in the nth time slot is obtained.

$$R^{cov}_{du,d_{rp_j}}[n] = W\log_2(1+\beta)\times\left(\frac{n_{us,d_{rp_j}}}{100}\right) \tag{56}$$

Algorithm 2 calculates the number of downlink users covered by UAV at the jth random point and nth time slot, coverage probability and available rate for downlink users.

Algorithm 2 Calculation of coverage probability and available rate for all points and DUs

1st step: set parameters $r_{u_i}[n], r_{rp_j}[n], \varphi_{u_i}[n], \varphi_{u_{rp_j}}[n]$, n_d^{min}, R_{max}
2nd step: check the condition $n_d \leqslant n_d^{min}$
3rd step: the condition is true → start the loop for parameters n, i, j and continue executing the commands (from the fifth step)
4th step: the condition is false → return to the condtion (second step)
5th step: calculate $d_{u_i,d_j}[n]$
6th step: labeling ε0ε for $z_{u_i,d_{rp_j}}[n]$ according to the condition $d_{u_i,d_j}[n] \leqslant R_{max}$
7th step: labeling ε1ε for $z_{u_i,d_{rp_j}}[n]$ according to the condition $d_{u_i,d_j}[n] > R_{max}$
8th step: calculate the number of DUs covered and store in parameter $N_{us,d_{rp_j}}[n]$
9th step: calculate $P^{cov}_{du,d_{rp_j}}[n]$
10th step: calculate $R^{cov}_{du,d_{rp_j}}[n]$
11th step: calculate $h^*[n] \leftarrow \left[\frac{\partial\bar{L}(R[n],h[n])}{\partial h[n]} = 0\right]$
12th step: return values $N_{us,d_{rp_j}}[n]$, $P^{cov}_{du,d_{rp_j}}[n]$, $R^{cov}_{du,d_{rp_j}}[n]$

In the first step of Algorithm 2, the position of downlink users in terms of polar coordinates and other information such as the maximum radius obtained from Algorithm 1 and the threshold of the number of D2D users from the UAV processor memory as input data of Algorithm 2 be. In the second step, a condition is applied to check the number of D2D users to prevent possible interference. Increasing the number of D2D transmitters, as we mentioned in the previous section, can lead to a decrease in the coverage probability and available rate. In the third step, if the condition is true, the upper and lower bounds for the time slots, the counter of downlink users and the counter of random points are applied for calculation. In the fourth step, if the condition is false, data processing resumes from the second step. In the fifth step, the distance of each user from random points and all time slots is calculated. In the sixth and seventh steps, the process of calculating user coverage or lack of coverage is defined. In this step, if the distance of the user is less than the value of R_{max}, the user is covered, otherwise, that is not covered. This is done by putting zero and one in $z_{u_i,d_{rp_j}}[n]$.

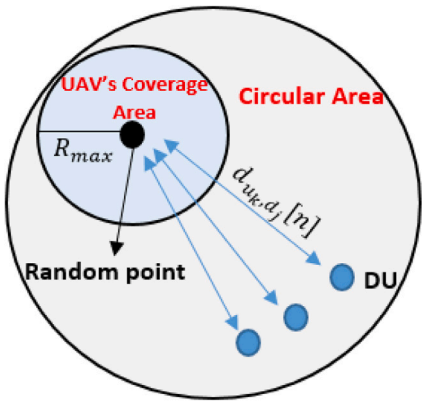

Fig. 8. The distance between DU and random points (it means the center of the area covered by the drone with the radius R_{max}).

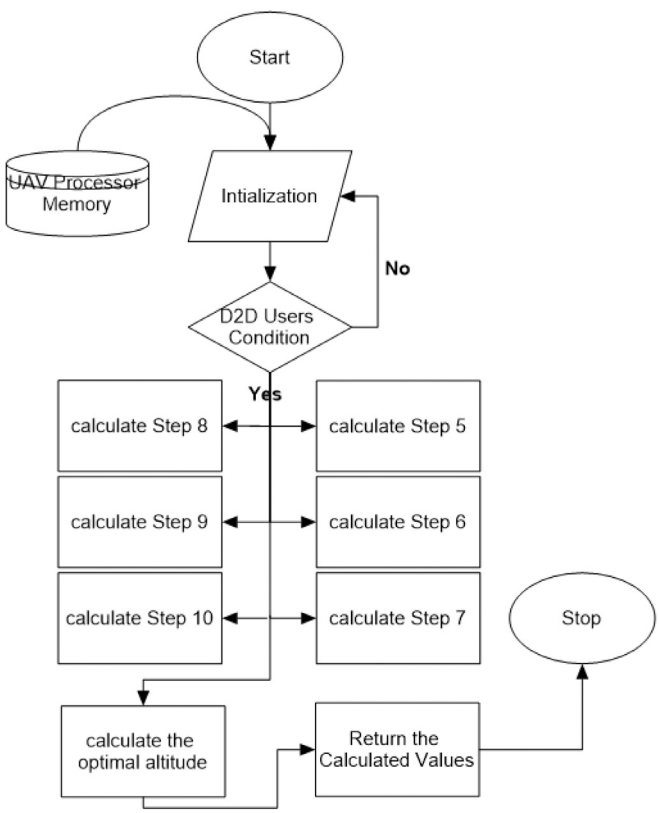

Fig. 9. Flowchart of Algorithm 2.

In the eighth step, the total number of users covered at each random point is calculated. In the 9th, 10th and 12th steps, the probability of coverage and the available rate for downlink users are calculated based on the Shannon channel capacity, respectively, and their values are stored in the UAV's processor memory for further processing. In the eleventh step, the optimal height of the UAV is calculated in each time slot. Fig. 9 shows the flowchart of algorithm 2.

2.2.3. Finding the optimal point of the UAV

In this section, we introduce an algorithm to select the optimal points of the UAV in each time slot. We note that, the optimal point of the drone is the point that brings the highest possible coverage and available rate for downlink users. If $P_M^*[n]$, the optimal point of the UAV is in the nth time slot. So we have:

$$P_M^*[n] = \arg\max_{P,R}\left(\vec{V}_M[n]\right), \vec{V}_M[n] = \begin{bmatrix} P_{cov,d_{rp_j}}[n] \\ R_{cov,d_{rp_j}}[n] \end{bmatrix}_{2\times 1}^{T} \tag{57}$$

Where, $\vec{V}_M[n]$ is a vector with dimensions of 2×1 and has two components including coverage probability and available rate for the downlink users. Eq. (57) is the maximum value problem for random points. To solve this problem, we use algorithm 3.

Algorithm 3 Calculation of the optimal point for the UAV

1st step: set parameters $P^{cov}_{du,d_{rp_j}}[n], R^{cov}_{du,d_{rp_j}}[n], \tau_1, \tau_2$
2nd step: filter the calculated set and data according to the conditions $P^{cov}_{du,d_{rp_j}}[n] \geqslant \tau_1, R^{cov}_{du,d_{rp_j}}[n] \geqslant \tau_2$
3rd step: Put the filtered sets in the $Filter1$ and $Filter2$ parameters
4th step: put the first value of the $Filter1$ set in the $temp1$ variable
5th step: put the first value of the $Filter2$ set in the $temp2$ variable
6th step: start the loop for parameters n, i, j
7th step: check the condition $Filter1 > temp1$ $\&\& Filter2 > temp2$
8th step: the condition is true → return the values of $temp1$ and $temp2$ and continue executing the commands (from the tenth step)
9th step: the condition is incorrect → return to the loop (seventh step)
10th step: Find the random points corresponding to parameters $temp1$, $temp2$ and placing in parameter $P_M^*[n]$
11th step: Exit from the loop

In the first step of the algorithm 3, initialization is done from the UAV's processor memory. In the second step, by setting the threshold on the average probability of coverage and the average available rate of downlink users, pre-processing is done to increase the speed of information processing and also to reduce the UAV's memory space. τ_1 and τ_2 are the average coverage probability threshold and available rate for downlink users, respectively. In the third, fourth and fifth steps, first, the filtered sets are stored in the $Filter1$ and $Filter2$ sets. $Filter1$ is the average filtered set of coverage probability for downlink users and $Filter2$ is the average filtered set of available rate for downlink users. After this operation, in order to start the process of finding the maximum value, the first value of the $Filter1$ set and the $Filter2$ set are stored in the $temp1$ and $temp2$ variables, respectively. In the sixth step, a loop is set for time slots, downlink users and random points to value from the lower bound to the upper bound. In the seventh step, the maximum value condition is applied to the average coverage probability and the average available rate for downlink users. In the eighth step, if the condition is true, the temporary values stored in $temp1$ and $temp2$ variables are returned. In the ninth step, if the condition is false, the algorithm returns to the seventh step. Finally, in the tenth step, the optimal points of the UAV are obtained in each time slot. Fig. 10 shows the flowchart of algorithm 3.

2.3. Analysis of computational complexity and optimality of algorithms 1, 2 and 3

If ℓ is the number of states, $T(\ell)$ is the computational complexity, and $\mathcal{O}(\ell)$ is the time complexity, the computational and time complexities for algorithms 1, 2 and 3 are obtained from relations (58), (59), and (60), respectively.

$$T_{AL1}(\ell) = 5\ell + 3 \rightarrow \mathcal{O}(\ell) \tag{58}$$

$$T_{AL2}(\ell) = \ell^3 + 8\ell + 7 \rightarrow \mathcal{O}(\ell^3) \tag{59}$$

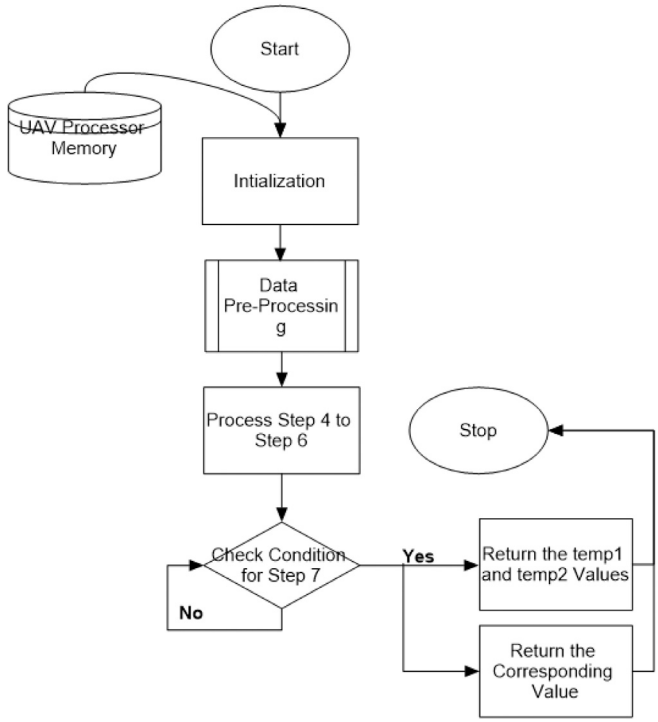

Fig. 10. Flowchart of Algorithm 3.

Table 1
Simulation parameters.

Description	Value	Parameter
UAV transmit power	5 W	P_u
D2D transmit power	100 mW	P_d
SINR threshold	2	β
Path loss coefficient	−30 dB	K
Path loss exponent for UAV-user link	2	α_d
Path loss exponent for D2D link	3	α_u
Noise power	−120 dBm	N
Bandwidth	1 MHz	W
D2D pair fixed distance	20 m	d_0
Excessive attenuation factor for NLoS	20 dB	η
Parameters for dense urban environment	0.136, 11.95	B, C
The radius of the circular area	1000 m	R_c
Threshold of R_{max}	0.9	ε
Number of downlink users	100	n_{du}
Number of D2D users	5	n_d
Maximum number of time slots	50	M
The minimum and maximum direction of movement of downlink users	$[0, 2\pi]$ rad	$[d_{min}, d_{max}]$
The minimum and maximum velocity of movement of downlink users	$[1, 10]\frac{m}{s}$	$[v_{min}, v_{max}]$
The highest speed of the UAV	$10\frac{m}{s}$	v_{uav}^{max}
Optimal Altitude [27]	500 m	h^*
Average additional loss for LoS link and dense urban environment [27]	1.6 dB	ξ_{LoS}
Average additional loss for NLoS link and dense urban environment [27]	23 dB	ξ_{NLoS}
Threshold of $P_{du,d_{rp_j}}^{cov}[n]$	0.9	τ_1
Threshold of $R_{du,d_{rp_j}}^{cov}[n]$	0.1	τ_2
The total flight time of the UAV	300 s	T_{uav}

$$T_{AL3}(\ell) = \ell^3 + 13\ell + 3 \rightarrow \mathcal{O}(\ell^3) \tag{60}$$

In relation (58), there are a total of 5ℓ states for executing commands and 3 calculation commands. Therefore, the computational complexity of algorithm 1 is $5\ell+3$ and its time complexity is $\mathcal{O}(\ell)$. In relation (59), a total of ℓ^3 states due to the existence of three loops to repeat n, i, j, a total of 8ℓ states in other steps to execute commands (for example, the execution of the condition of the second step which has ℓ states and...) and There are a total of 7 calculation commands. Therefore, the computational complexity of algorithm 2 is $\ell^3 + 8\ell + 7$ and its time complexity is $\mathcal{O}(\ell^3)$. In relation (60), there are a total of ℓ^3 states due to the existence of three loops for repeating n, i, j, a total of 13ℓ states in other steps to execute commands, and a total of 3 calculation commands. Therefore, the computational complexity of algorithm 3 is $\ell^3+13\ell+3$ and its time complexity is $\mathcal{O}(\ell^3)$. Considering that ℓ^3 is the largest state among the three algorithms, therefore the time complexity for algorithms 1, 2, and 3 is $\mathcal{O}(\ell^3)$.

To prove the optimality of the proposed algorithm, we use asymptotic analysis. In asymptotic analysis, the growth rate of functions are compared with each other. According to the definition of equivalence relation, two functions f and g are asymptotically equivalent ($f \sim g$ or $g \sim f$), if we have:

$$f \sim g = \lim_{n\to\infty} \frac{f}{g} = 1 \tag{61}$$

According to (61), if the division limit of two functions equals one at infinity, then those two functions are asymptotically equivalent. Otherwise, they are non-equivalent. We use (61) to compare the time complexity of the proposed algorithm and other methods. According to Ref. [15] in paper, the time complexity of the presented algorithm is $\mathcal{O}(\ell^7)$ according to the number of iterations and loops of the algorithm for the relaxation of the optimization problem and the related relationship up to 7 steps. According to (61), we calculate the asymptotic analysis for the proposed algorithm and the algorithm presented in [15], we have:

$$\mathcal{O}(\ell^7) = \ell^7, \mathcal{O}(\ell^3) = \ell^3 \rightarrow \lim_{\ell\to\infty} \frac{\ell^7}{\ell^{\ell^3}} = \infty, \lim_{\ell\to\infty} \frac{\ell^3}{\ell^{\ell^7}} = 0 \rightarrow \ell^7 > \ell^3 \tag{62}$$

From (62), it is clearly seen that the growth rate of function ℓ^7 is higher than that of function ℓ^3. Therefore, it can be concluded that the proposed algorithm with a time complexity of $\mathcal{O}(\ell^3)$ has a lower execution time than the algorithm presented with a complexity of $\mathcal{O}(\ell^7)$ in [15].

In [17], the time complexity for the Q-Learning method is $\mathcal{O}(\ell^{\mathscr{g}})$ and $\mathscr{g} = 4$. It is clear that the proposed algorithm with a time complexity of $\mathcal{O}(\ell^3)$ has a lower execution speed than the [17] with a time complexity of $\mathcal{O}(\ell^4)$ ($\mathcal{O}(\ell^4) > \mathcal{O}(\ell^3)$). The comparison of the asymptotic analysis for the proposed algorithm and the algorithm presented in [17] is given in relation (63).

$$\mathcal{O}(\ell^4) = \ell^4, \mathcal{O}(\ell^3) = \ell^3 \rightarrow \lim_{\ell\to\infty} \frac{\ell^4}{\ell^{\ell^3}} = \infty, \lim_{\ell\to\infty} \frac{\ell^3}{\ell^{\ell^4}} = 0 \rightarrow \ell^4 > \ell^3 \tag{63}$$

According to [18], the time complexity of the proposed algorithm is $\mathcal{O}(\ell^5)$. According to (61), we calculate the asymptotic analysis for the proposed algorithm and the algorithm presented in [18], we have:

$$\mathcal{O}(\ell^5) = \ell^5, \mathcal{O}(\ell^3) = \ell^3 \rightarrow \lim_{\ell\to\infty} \frac{\ell^5}{\ell^{\ell^3}} = \infty, \lim_{\ell\to\infty} \frac{\ell^3}{\ell^{\ell^5}} = 0 \rightarrow \ell^5 > \ell^3 \tag{64}$$

From [18], it is clearly seen that the growth rate of function ℓ^5 is higher than that of function ℓ^3. Therefore, it can be concluded that the proposed algorithm with a time complexity of $\mathcal{O}(\ell^3)$ has a lower execution time than the algorithm presented with a complexity of $\mathcal{O}(\ell^5)$ in Ref. [18].

3. Simulation results and analysis

The simulation of this article has been done in MATLAB environment. The simulation parameters of this article are set according to Table 1. Some of these parameters are valued based on Ref. [26].

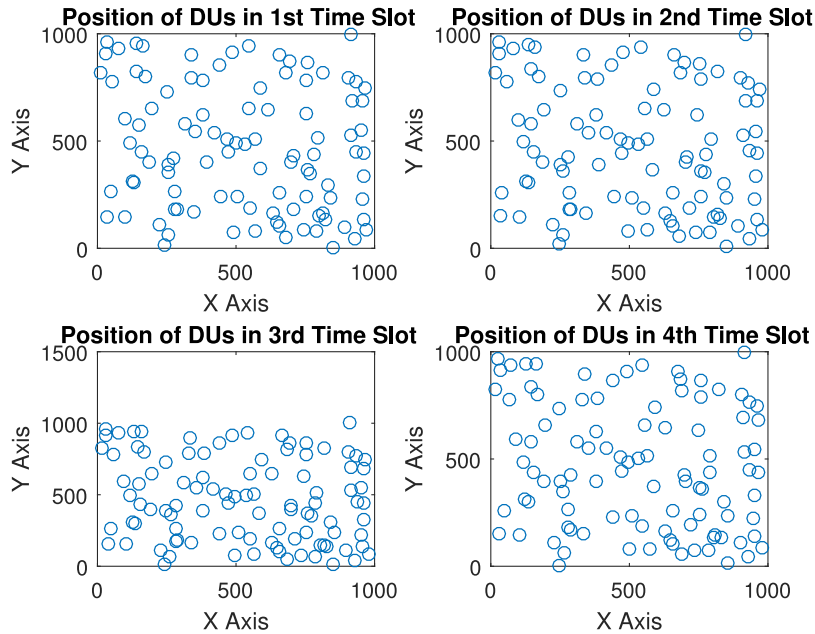

Fig. 11. Downlink users position for the first 4 time slots.

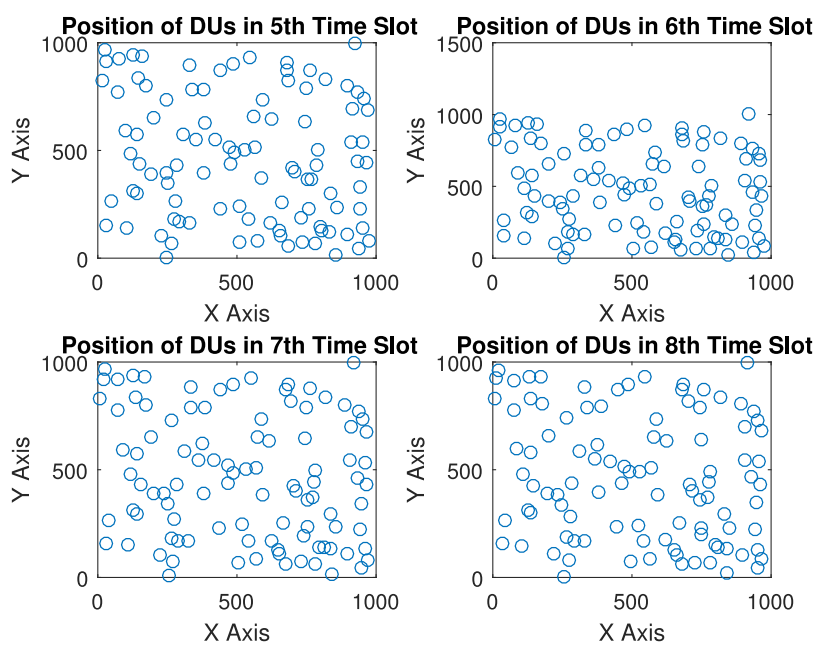

Fig. 12. Downlink users position for the second 4 time slots.

Figs. 11, 12 and 13, show the position of downlink users for the first, second and third 4 time slots for the movement interval $x, y = [1, 1000]$ in the Cartesian coordinate system, respectively.

In Fig. 11 and in the first time slot, the users are uniformly distributed in the area. Then in the next time slots, each user starts moving according to the Gauss–Markov random model. According to Table 1, each user is assigned a random speed and direction in a specified interval. Then, the new speed and direction for the next time slot are calculated by (8) and (9) for each user. In this way, users move up to the 50th time slot.

Fig. 14 shows the average coverage probability in terms of the radius of the area covered by the UAV. According to this figure, the coverage probability had an upward trend at first, that is, up to a radius of 647 m, then it went downward and reached a constant value. This figure has been obtained according to Algorithm 1, for the optimal height of 500 m, radius value for 1 to 4500 meters (lower and upper bounds) and threshold $\epsilon = 0.9$ for the probability of covering the downlink users. According to the mentioned explanations, the maximum radius of the area covered by the UAV is calculated to be 647 m.

Fig. 15 shows the number of downlink users served per 100 random points and for 50 time slots by Algorithm 2. This figure specifies, for each point, how many downlink users are covered by the drone in different time slots. As can be seen in the figure, the changes for different time slots are small, and only a few time slots show a lot of changes. This is due to the mobility model and the radius of the circular region. It is normal that these measurements are affected by increasing the radius of the circular area and changing the mobility model.

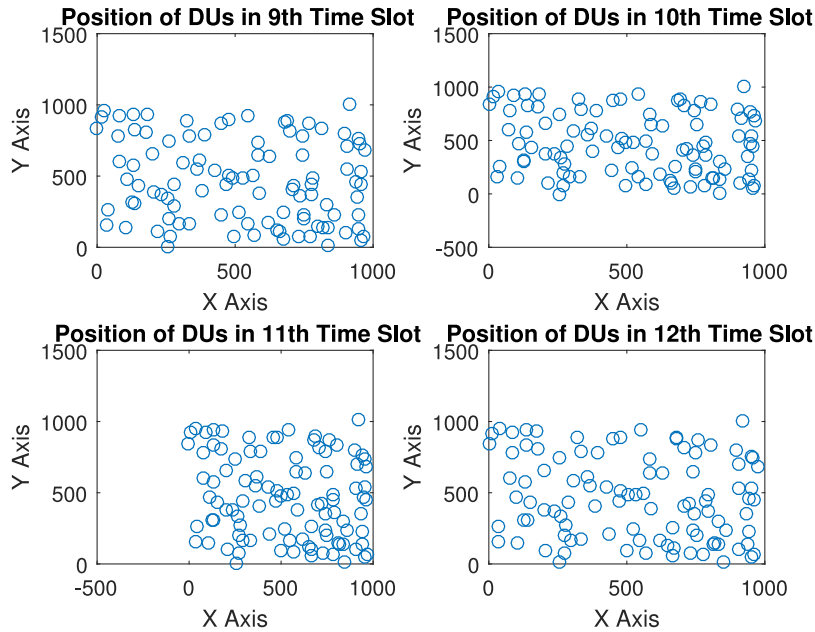

Fig. 13. Downlink users position for the third 4 time slots.

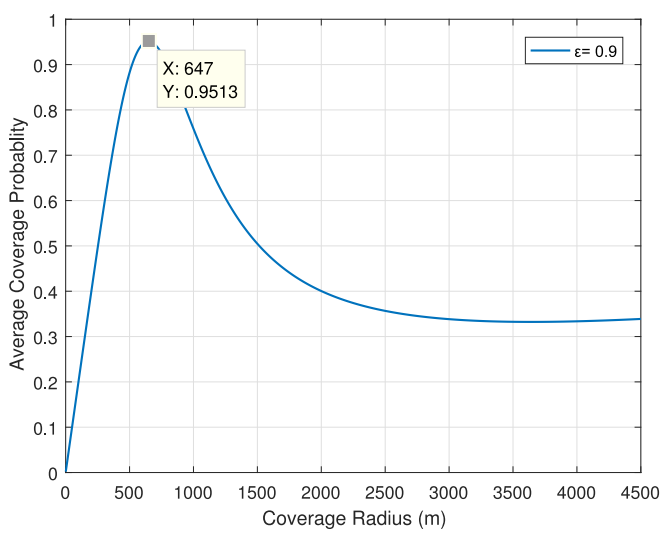

Fig. 14. Average coverage probability according to the radius of the area covered by the UAV.

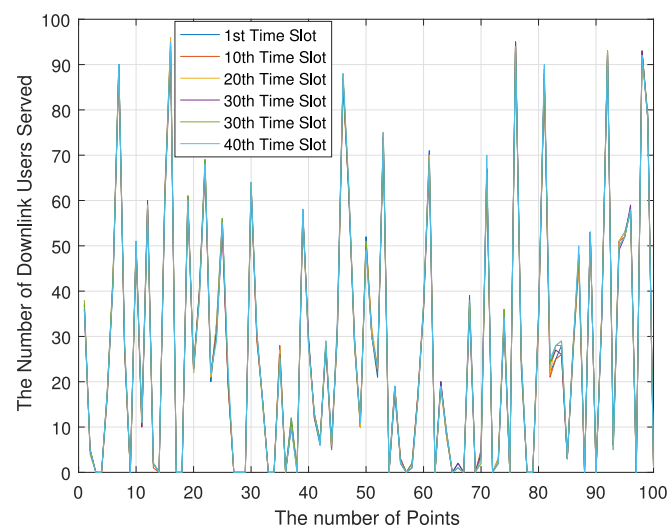

Fig. 15. The total number of downlink users served for 100 random points in different time slots.

Fig. 16 shows the optimal points where the UAV should be present in each time slot based on Algorithm 3. In the mentioned figure, for the first time slot, the 76th random point is considered as the optimal point. Then the drone moves and goes to the 16th point for the second and third time slots. Again, it returns to the 76th random point in the third time slot. This process continues until the 50th time slot. The total number of UAV trips between the 16th and 76th points is ten times.

Fig. 17 shows the position of the optimal points of the UAV among 100 random points. The red points are the points where the UAV is deployed in 50 time slots according to Fig. 16 and provides service to moving downlink users.

Fig. 18 shows the number of downlink users served in different methods and the proposed method from the first to the seventh time slot. The reason for the comparison for 7 time slots is that only the results of 7 time slots are available from other methods. If according to the simulation parameters, we have simulated for 50 time slots. According to the obtained results, the proposed method serves the largest number of downlink users among other methods. In the 3D deployment method, on average, 18.6508 users out of 100 users are covered. This method provides the lowest amount of coverage among other methods. In the trajectory design method, on average, 51.2897 users out of 100 users are served. This method covers more users than the 3D deployment method, but it does not perform well compared to the trajectory optimization method. In the trajectory optimization method, an average of 61.7019 users out of 100 users are covered. Among other methods (apart from the proposed method), this method provides the highest amount of services to users. We note that these results were obtained considering a single UAV. More details of this comparison are given in Table 2 percentage of error is defined as $E = \frac{(ApproximateValue - ExactValue)}{ExactValue} \times 100$.

The reason for the superiority of the proposed method over other methods is the measurement of two factors (average available rate and coverage probability) for all points in the target area, checking the entire area and choosing the best point. This will reduce the percentage

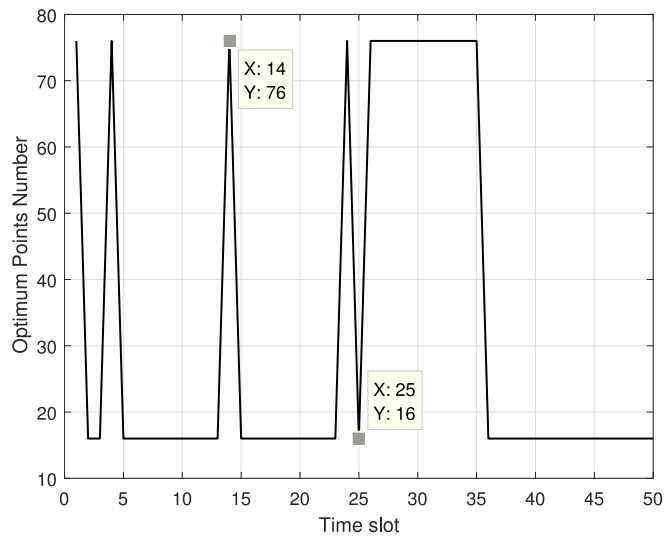

Fig. 16. UAV optimal points for fifty time slots.

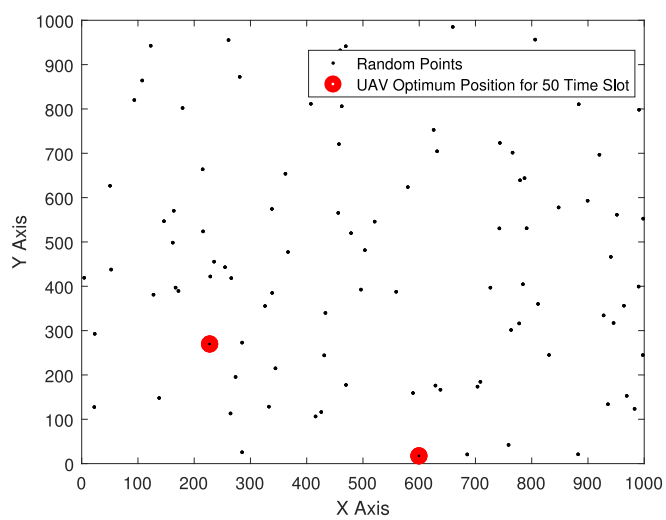

Fig. 17. Optimal points selected for UAV in 50 time slots from among all random points.

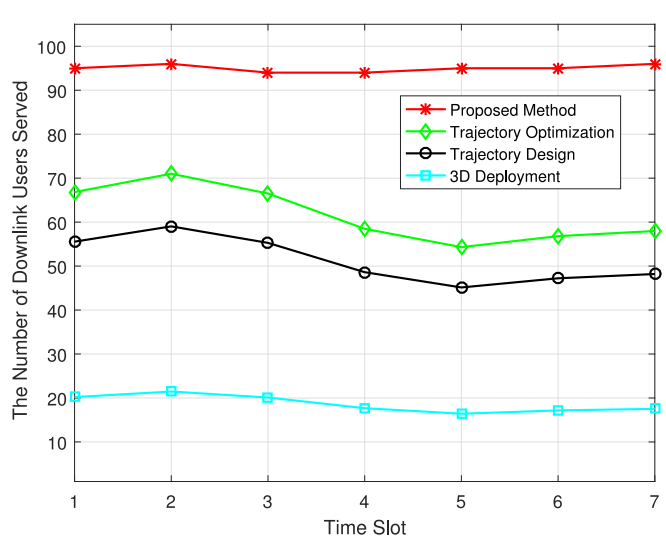

Fig. 18. The number of downlink users served in the proposed method and other methods for the first 7 time slots.

Table 2
Simulation parameters.

Percentage of error	Real-Time	GPS	Single UAV	Method
81.342	✓	✓	18.6508	3D Deployment [18]
48.7103	×	✓	51.2897	Trajectory Design [17]
38.2981	×	✓	61.7019	Trajectory Optimization [15]
5.1400	✓	✓	94.86	Proposed Method

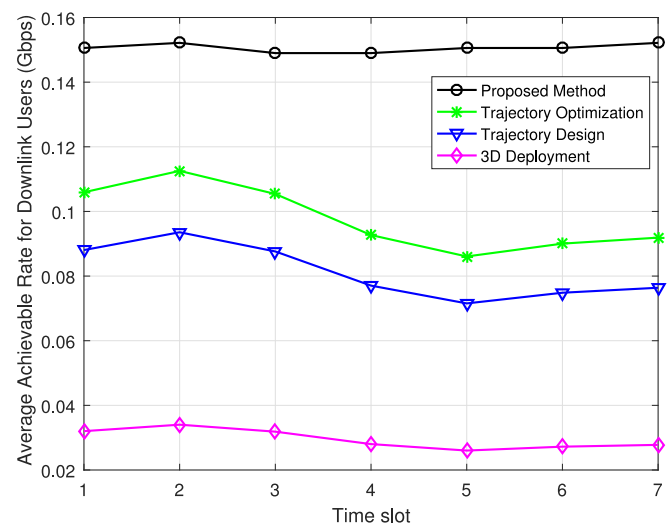

Fig. 19. The average available rate for downlink users in the proposed method and other methods for the first 7 time slots.

of errors and provide the most possible coverage for users. We note that in non-real-time scenarios, increasing the speed and volume of information processing to calculate the optimal point of the UAV does not create sensitivity.

Fig. 19 shows the average available rate for downlink users in different methods and the proposed method for the first to seventh time slots. The output of this figure and Fig. 20 are almost identical in appearance, because the coverage probability is a factor of the channel capacity and has a direct relationship with the average available rate. Among the proposed methods for bandwidth $W = 1$ MHz and the threshold of signal to interference plus noise ratio to $\beta = 2$, the proposed method has the highest average available rate for downlink users (according to the mathematical relationship of average available rate) provide. The average available rate is calculated to be 0.0296 Gbps on average in the 3D deployment method. Compared to other methods, this method provides the lowest rates for downlink users. The average available rate in the trajectory design method is 0.0813 Gbps. This method provides a higher rate for downlink users than the 3D deployment method. The average available rate is calculated as 0.0978 Gbps in the trajectory optimization method. This method provides a higher rate for downlink users than the trajectory design and 3D deployment methods. The average available rate in the proposed method is 0.1506 Gbps. In total, the proposed method provides the highest rate for Feroso users. In Fig. 20, average values of path loss per 1 to 4500 meter are measured for altitude. Also, the radius of the area covered by the UAV for simulation is 500 m. As can be seen from the figure, in the initial intervals of altitude, the average path loss decreased and then increased. According to the figure and (31), the height should be adjusted in such a way as to prevent the increase of path losses (transmitted power).

Fig. 21 shows the average path loss function as a function of UAV height for R values from 100 m to 1000 m. In Fig. 21, it can be seen

Table 3
Example of calculation of optimal height by algorithm 2 (for R from 100 m to 1000 m)

R	100 m	200 m	300 m	400 m	500 m	600 m	700 m	800 m	900 m	1000 m
h^*	96 m	193 m	289 m	386 m	482 m	578 m	675 m	770 m	868 m	964 m

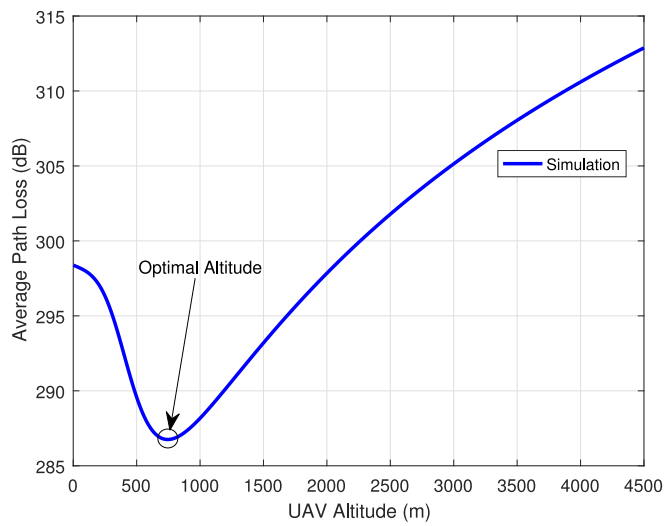

Fig. 20. The average path loss versus UAV's altitude.

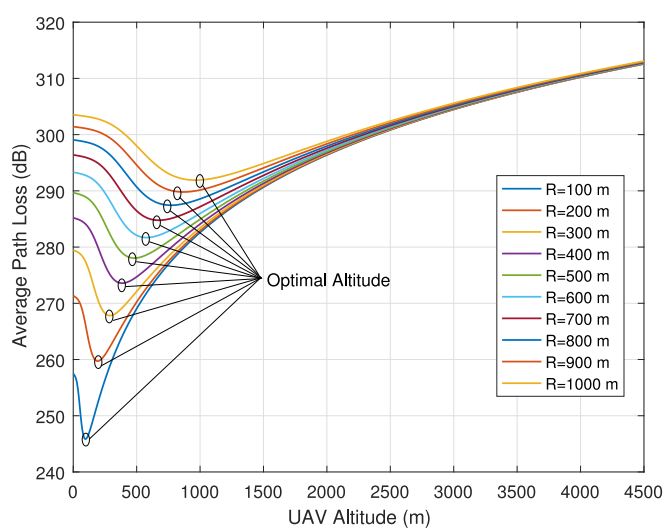

Fig. 21. The average path loss function according to the height of the drone (for R from 100 meters to 1 km).

that the optimal height of the UAV is at the minimum of the average path loss function. Therefore, by deriving the average path loss function with respect to the height and setting it equal to zero (according to (59)), the optimal height of the UAV can be calculated for the value of R. The eleventh step of algorithm 2 is responsible for calculating the optimal height of the UAV in each time slot. The details of Fig. 2 and the example of calculating the optimal height in each time slot by Algorithm 2 are given in Table 3. Fig. 22 shows the algorithm execution time for the proposed algorithm and other algorithms in terms of input size. This comparison is measured per 1000 inputs for each algorithm. As seen in this figure, the highest execution time is related to the algorithm used in the trajectory optimization method and the lowest

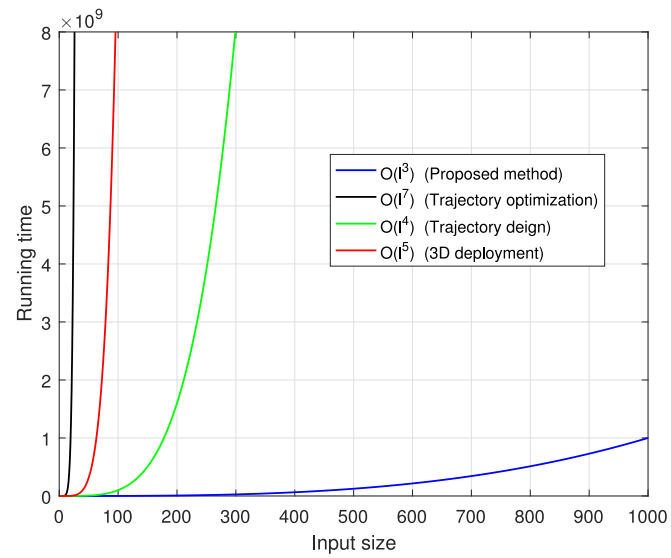

Fig. 22. The execution time of the proposed algorithm and other algorithms according to the size of the input.

execution time is related to the proposed method ($\mathcal{O}(\ell^7) < \mathcal{O}(\ell^5) < \mathcal{O}(\ell^4) < \mathcal{O}(\ell^3)$).

4. Conclusion

In this paper, we considered a moving single UAV as a flying base station in an area where the device-to-device transmitter–receiver pair has D2D communication. In this way, two types of users, including Downlink users with Gauss–Markov random mobility served by UAV and D2D transmitters who are directly connected with their corresponding receiver with a fixed position, are considered as the system model. According to the objectives of the article, in order to find the optimal position of the UAV, we first considered a number of random points in the circular area and by calculating the distance between the points and the users using the proposed method, we calculated the best point of the UAV for each time slot, which is the most It provides the number of serviced users (average coverage probability) and the available rate for DUs compared to other methods according to algorithm 3.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Proof of (25)

We assume that the two random variables X_{rp} and Y_{rp}, which are the positions of random points in the Cartesian coordinate system,

are uniformly distributed on the circle D_r and with the radius R_c. Therefore, D_r is obtained as (A.1).

$$D_r = \{(x_{rp}, y_{rp}) \in \mathbb{R}^2 | x_{rp}^2 + y_{rp}^2 \leqslant R_c\} \tag{A.1}$$

The joint cumulative distribution function (CDF) for the location of random points in the polar coordinate system is obtained by the (A.2)

$$\left(r_{rp} = \sqrt{x_{rp}^2 + y_{rp}^2}, \varphi_{rp} = \tan^{-1}\left(\frac{y_{rp}}{x_{rp}}\right)\right).$$

$$\begin{aligned} F_{R_{rp},\varphi_{rp}}(r_{rp}, \varphi_{rp}) &= P(R_{rp} < r_{rp} \cap \phi_{rp} < \varphi_{rp}) \\ &= \frac{A_{sec}}{\pi R_c^2} = \frac{r_{rp}^2 \frac{\varphi_{rp}}{2\pi}}{R_c}, 0 < r_{rp} < R_c, 0 < \varphi_{rp} < 2\pi \end{aligned} \tag{A.2}$$

Where, A_{sec} is the area of the sector of a part of the circle in the specified area. By taking the derivative of (A.3) with respect to the parameters of the polar coordinate system, the joint probability density function is obtained.

$$\begin{aligned} f_{R_{rp},\varphi_{rp}}(r_{rp}, \varphi_{rp}) &= \frac{\partial^2}{\partial r_{rp} \partial \varphi_{rp}} F_{R_{rp},\varphi_{rp}}(r_{rp}, \varphi_{rp}) \\ &= \begin{cases} \frac{r_{rp}}{\pi R_c^2} & 0 \leqslant r_{rp} \leqslant R_c, 0 \leqslant \varphi_{rp} \leqslant 2\pi \\ 0 & \text{Otherwise} \end{cases} \end{aligned} \tag{A.3}$$

And thus (25) is proved.

Appendix B. Proof of (48) and (49)

The coverage probability for a cellular user located in (r, φ) is defined by (B.1) [26].

$$\begin{aligned} P_{cov,du}(r, \varphi, \beta) = \mathbb{P}[\gamma_u \geqslant \beta] &= P_{LoS}(r)\mathbb{P}\left[\frac{P_u r^{-\alpha_u}}{I_d + N} \geqslant \beta\right] \\ &+ P_{NLoS}(r)\mathbb{P}\left[\frac{\eta P_u r^{-\alpha_u}}{I_d + N} \geqslant \beta\right] \\ &= P_{LoS}(r)\mathbb{P}\left[I_d \leqslant \frac{P_u r^{-\alpha_u} - \beta N}{\beta}\right] \\ &+ P_{NLoS}(r)\mathbb{P}\left[I_d \leqslant \frac{\eta P_u r^{-\alpha_u} - \beta N}{\beta}\right] \end{aligned} \tag{B.1}$$

We know that there is no closed-form expression for the CDF of D2D user interference. Here, we present upper and lower bounds for the interference CDF. First, D2D transmitters with interference are divided into two subsets in the form of relation (B.2) [26].

$$\begin{cases} \phi_1 = \{\phi_B | P_d d_i^{-\alpha_d} g_i \geqslant T\} \\ \phi_2 = \{\phi_B | P_d d_i^{-\alpha_d} g_i \leqslant T\} \end{cases} \tag{B.2}$$

Where, T is a threshold to obtain the CDF of interference of D2D users. Now, according to the interference power of D2D users located in ϕ_1 and ϕ_2 as I_{d,ϕ_1} and I_{d,ϕ_2}, Eq. (B.3) can be written [26].

$$\begin{aligned} \mathbb{P}[I_d \leqslant T] &= \mathbb{P}\left[I_{d,\phi_1} + I_{d,\phi_2} \leqslant T\right] \leqslant \left[I_{d,\phi_1} \leqslant T\right] \\ &= \mathbb{P}[\phi_1 = 0] = \mathbb{E}\left[\prod_{\phi_B} \mathbb{P}\left(P_d d_i^{-\alpha_d} g_i < T\right)\right] \\ \mathbb{E}\left[\prod_{\phi_B} \mathbb{P}\left(g_i < \frac{T d_i^{\alpha_d}}{P_d}\right)\right] &\overset{(a)}{=} \mathbb{P}\left[\prod_{\phi_B} 1 - exp\left(-\frac{T d_i^{\alpha_d}}{P_d}\right)\right] \\ &\overset{(b)}{=} exp\left(-\lambda_d \int_0^\infty exp\left(-\frac{T r^{\alpha_d}}{P_d}\right) r dr\right) \\ &= exp\left(-\pi\lambda_d \left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}} \Gamma\left(1 + \frac{2}{\alpha_d}\right)\right) \end{aligned} \tag{B.3}$$

Where, (a) and (b) are obtained by assuming Rayleigh fading and PPP probability generating functional (PGFL). The upper bound is obtained in the form of (B.4) [26].

$$\begin{aligned} \mathbb{P}[I_d \geqslant | \phi_1 = 0] &\overset{(a)}{\leqslant} \frac{\mathbb{E}[I_d \geqslant T \mid \phi_1 = 0]}{T} \\ &= 1 - \Big(\mathbb{P}\left[I_d \geqslant T \mid I_{d,\phi_1} \geqslant T\right]\mathbb{P}\left[I_{d,\phi_1} \geqslant T\right] \\ &\quad + \mathbb{P}\left[I_d \geqslant T \mid I_{d,\phi_1} \leqslant T\right]\mathbb{P}\left[I_{d,\phi_1} \leqslant T\right]\Big) \\ &= 1 - \Big(\mathbb{P}\left[I_{d,\phi_1} \geqslant T\right] + \mathbb{P}\left[I_d \geqslant T \mid I_{d,\phi_1} \leqslant T\right] \\ &\quad \times \mathbb{P}\left[I_{d,\phi_1} \leqslant T\right]\Big) \\ &= 1 - \Big(1 - \mathbb{P}[\phi_1 = 0] + \mathbb{P}\left[I_d \geqslant T \mid I_{d,\phi_1} \leqslant T\right] \\ &\quad \times \mathbb{P}[\phi_1 = 0]\Big) \\ &= \mathbb{P}[\phi_1 = 0]\left(1 - \mathbb{P}\left[I_d \geqslant T \mid \phi_1 = 0\right]\right) \end{aligned} \tag{B.4}$$

Also, we have [26]:

$$\begin{aligned} \mathbb{P}[I_d \geqslant T \mid \phi_1 = 0] &\overset{(a)}{\leqslant} \frac{\mathbb{E}[I_d \geqslant T \mid \phi_1 = 0]}{T} \\ &= \frac{1}{T}\mathbb{E}\left[\sum_{\phi} P_d d_i^{-\alpha_d} g_i \mathbf{1}\left(P_d d_i^{-\alpha_d} g_i \leqslant T\right)\right] \\ &= \frac{1}{T}\mathbb{E}_{d_i}\left[\sum_{\phi} P_d d_i^{-\alpha_d} \mathbb{E}\left[g_i \mathbf{1}\left(g_i \leqslant \frac{T d_i^{-\alpha_d}}{P_d}\right)\right]\right] \\ &= \frac{1}{T}\mathbb{E}_{d_i}\left[\sum_{\phi} P_d d_i^{-\alpha_d}\left[\int_0^{\frac{T d_i^{\alpha_d}}{P_d}} g e^{-g} dg\right]\right] \\ &= \frac{2\pi P_d \lambda_d}{T}\int_0^\infty r^{-\alpha_d}\left(\int_0^{\frac{T d_i^{\alpha_d}}{P_d}} g e^{-g} dg\right) r dr \\ &= \frac{2\pi\lambda_d \Gamma\left(1 + \frac{2}{\alpha_d}\right)}{\alpha_d - 2}\left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}} \end{aligned} \tag{B.5}$$

Section (a) is based on Markov's inequality, which is stated as follows: For any non-negative integrable random variable X and positive L, $\mathbb{P}(X \geqslant L) \leqslant \frac{\mathbb{E}[X]}{L}$. Also, $\mathbf{1}(.)$ is an indicator function that can only be equal to zero or one. Therefore, the lower (L_I) and upper (U_I) bounds for the cumulative interference distribution function are obtained by (B.6) and (B.7), respectively.

$$\begin{aligned} L_I(T) = &\left[1 - \frac{2\pi\lambda_d \Gamma\left(1 + \frac{2}{\alpha_d}\right)}{\alpha_d - 2}\left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}}\right] \\ &\times exp\left(-\pi\lambda_d \left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}} \Gamma\left(1 + \frac{2}{\alpha_d}\right)\right) \end{aligned} \tag{B.6}$$

$$U_I(T) = \times exp\left(-\pi\lambda_d \left(\frac{T}{P_d}\right)^{\frac{-2}{\alpha_d}} \Gamma\left(1 + \frac{2}{\alpha_d}\right)\right) \tag{B.7}$$

Therefore, $L_I(T) \leqslant \mathbb{P}\{I_d \leqslant T\} \leqslant U_I(T)$. Finally, the lower and upper bounds of the average coverage probability for downlink users in the cell can be obtained by (B.8) and (B.9), respectively [26].

$$\begin{aligned} \tilde{P}_{cov,du}^L(\beta, h) = &\int_0^{R_c} P_{LoS}(r, h) L_I\left(\frac{P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2} dr \\ &+ \int_0^{R_c} P_{NLoS}(r, h) L_I\left(\frac{\eta P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2} dr \end{aligned} \tag{B.8}$$

$$\begin{aligned} \tilde{P}_{cov,du}^U(\beta, h) = &\int_0^{R_c} P_{LoS}(r, h) U_I\left(\frac{P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2} dr \\ &+ \int_0^{R_c} P_{NLoS}(r, h) U_I\left(\frac{\eta P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2} dr \end{aligned} \tag{B.9}$$

And thus relations (48) and (49) are proved.

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